15.4.2 CLAUSES IN SQL

SQL provides various clauses that can be used in the SELECT statements. This clauses can be called through python script. Almost all clauses will work with SQLite. The following frequently used clauses are discussed here

- DISTINCT
- WHERE
- GROUP BY
- ORDER BY.
- HAVING

15.4.2.1 SQL DISTINCT CLAUSE

The distinct clause is helpful when there is need of avoiding the duplicate values present in any specific columns/table. When we use distinct keyword only the unique values are fetched. In this example we are going to display the different grades scored by students from "student table".

Example 15.4.2.1-1

```
import sqlite3
connection = sqlite3.connect("Academy.db")
cursor = connection.cursor()
cursor.execute("SELECT DISTINCT (Grade) FROM student")
result = cursor.fetchall()
print(result)
```

OUTPUT

```
[('B',), ('A',), ('C',), ('D',)]
```

Without the keyword "distinct" in the above example displays 7 records instead of 4, since in the original table there are actually 7 records and some are with the duplicate values.

15.4.2.2 SQL WHERE CLAUSE

The WHERE clause is used to extract only those records that fulfill a specified condition. In this example we are going to display the different grades scored by male students from "student table"